Nicolaas Defaux

Senior Software Engineer · GenAI & Agentic Systems · AWS Cloud

Day rate: from £650 · Available: 1 month notice · Working: London / Remote / Hybrid · SC Cleared

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ABOUT

Lead software engineer who ships production AI systems end-to-end, from architecture through deployment and metrics. Five years building and rebuilding generative AI platforms for public-sector clients at IBM Consulting, including the first successful GenAI platform delivered for that client: six production releases, zero rollbacks, full NFR-suite adherence. I own observability, async-state architecture, and accessibility compliance, and I design and down the roadmap. Outside client work: freelance web builds and solo production apps to support my work and lifestyle, shipped design-to-launch.

TECHNICAL STACK

Languages & runtimes: TypeScript, JavaScript (Node.js, Express, Next.js, React, Jest), Python

AI/ML: AWS Bedrock, RAG, Bedrock Guardrails, AWS Agent Core, LLM integration & agents, Claude developer certified

Data: DocumentDB, PostgreSQL

Cloud & infra: AWS, Docker, Kubernetes, Dynatrace, Grafana, Terraform

Architecture: Microservices, RAG, agentic, event-driven (SSE), secure-by-design, hybrid cloud

Frontend & UX: Next.js, React, Accessibility (WCAG), GDS

CI/CD: GitLab, Helm

EXPERIENCE

IBM Consulting - *Tech Lead – current engagement* *2024 – Present.*

Hands-on across the agentic rebuild that becomes the reusable platform layer for the client's next generation of AI products by designing, scrutinising, and leading the build, not just directing it. Personally, converting bespoke, single-client builds into agentic components that other engagements can pull directly into production; merging a suite of sister services into a strategic, central agent-driven product. Own delivery end-to-end: AWS architecture, secure-by-design implementation, and the tooling that ships it.

Plagiarism Detection Tool - *Tech Lead · Public Sector Client* *2026*

Tech Lead on a plagiarism detection tool: through the early phase, I am personally driving solution design and refinement, applying public sector standards and data science approaches to take it from concept to a production-ready build. By leveraging my view of the GenAI estate, I am strategising a design and build with agentic workflow interoperability.

Generative AI for Learning (GAIL) - *Lead Engineer* - *Public Sector Client* *2024 – 2026*

Personally accountable for architecture and delivery of a Generative AI MVP that shipped to production. This was the first successful GenAI platform delivered for this client. Owned six production releases end-to-end with zero rollbacks. Built the observability layer (Dynatrace, Grafana) and the concurrent async-state architecture behind a multi-tab authoring UI; also delivered a GDS-compliant, accessible error-handling component and Kubernetes deployment diagnostics. Advanced systems thinking across a portfolio to acknowledge new-age Gen AI technology stack limitations and revise NFR specifications. Now personally leading GAIL's decommissioning; tearing down Kubernetes namespaces across every environment, destroying the AWS infrastructure via Terraform, and preparing components for open-source release. Full lifecycle ownership from inception to sunset, saving the parts that ship well.

Stack: *Next.js, TypeScript, Python · AWS Bedrock (Llama), RAG, Bedrock Guardrails · DocumentDB · EKS · Terraform · Dynatrace, Grafana*

Apply for a National Insurance Number (NINO) Service – *Associate Engineer - Public Sector Client* *2022 – 2023*

Shipped a citizen-facing, PII-handling service against a COVID-driven deadline, replacing an antiquated paper-driven process at speed. Delivered the technical business analysis and a hands-on development component personally.

Stack: *Next.js, TypeScript, Express.js, Jest, Docker, EKS, AWS*

Generative AI Lighthouse – Centre of Excellence - *Public Sector Client*

8 proofs-of-concept delivered, and 2 MVPs shipped for public-sector clients evaluating generative AI – bespoke AWS cloud architectures, foundational software, and GPT-4-powered showcases, plus an AI ethics framework and strategic AI adoption pipeline built to guide follow-on delivery.

Citizen Interaction History (R&D) - *Technical Architect* *Jan 2022 – Apr 2022*

Led requirement elicitation with client business and architectural stakeholders; the business case and recommendations produced led directly to onward contract procurement.

INDEPENDENT & FREELANCE DELIVERY

MAK99 Interiors – Studio Site Rebuild - *Freelance client* *2026*

Full Next.js rebuild of a Washington, D.C. interior design studio's marketing site, delivered end-to-end as billed freelance work: filterable project galleries with lightbox carousels, client testimonials, an Instagram feed, and SEO groundwork (JSON-LD, OG cards, sitemap). Live in production.

Stack: *Next.js, TypeScript, Tailwind CSS, Vercel*

World Cup 2026 Sweepstakes Tracker - *Solo build* *2026*

Production sweepstakes platform for the 2026 World Cup – live brackets, draw management, and leaderboards updating in real time – designed, built, and shipped solo for real players to use through the tournament.

Stack: *Next.js, TypeScript, Render, Neon, Cloudflare R2*

Seqflow - *Solo build* *Ongoing*

DSL-driven single source of truth for sequence diagrams, architecture inference, interface definitions, and RDS schema management – designed to live alongside code from the start of development. Hybrid text/visual canvas, message-contracts modelling, exports to OpenAPI and AsyncAPI.

EDUCATION

MSc Digital Technology Solutions Specialist – Northeastern University · First · 2024 – 2026

BSc Biology – University of York · 2:1 · 2016 – 2019

Software Engineering Apprenticeship (Level 5) – Distinction · 2021 – 2023